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Bromotrifluoromethoxylation of allenes: expedient access to allylic trifluoromethoxy derivatives†

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A mild and efficient method for the bromotrifluoromethoxylation of allenes has been explored using trifluoromethyl arylsulfonate (TFMS) as the trifluoromethoxy source. A series of allylic trifluoromethoxy derivatives were obtained with high yields and good regioselectivity by terminal selectivity of the allenes. This transition metal-free process expands the scope of the application of trifluoromethoxy reagents.

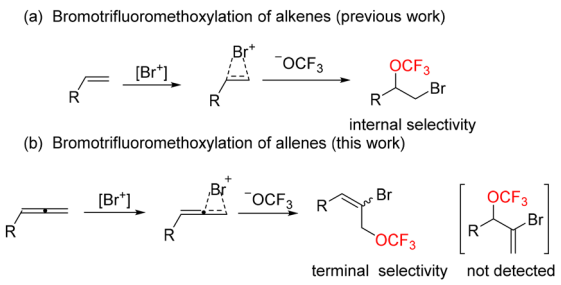
Introduction

The introduction of fluorinated functional groups into organic molecules has a significant impact on their physical and chemical properties.[1] Thus, more and more fluorinated drug molecules are being synthesized.[2] In particular, the trifluoromethoxy group (OCF_3) brings good metabolic stability and high lipophilicity to molecules, earning it the nickname of superhalogen.[3] The unique physical and chemical properties of trifluoromethoxy functional groups have attracted attention in the fields of materials science and pharmaceuticals. However, the number of marketed drugs and agricultural chemicals containing OCF_3 remains low, and the study of OCF_3 is largely behind the development of other fluorinated functional groups. Pioneering works have focused on the construction of the O–CF_3 bond from an already installed OH group, but this indirect method usually requires harsh conditions or toxic reagents (*e.g.*, HF and SF_4).[4] Directly introducing an OCF_3 functional group is an effective alternative method for obtaining trifluoromethoxylated compounds.[5] However, this is limited by the reversible decomposition of trifluoromethoxy and trifluoromethoxylation reagents.

In 2017, our research group developed trifluoromethyl benzenesulfonate (TFMS) as a trifluoromethoxy reagent, which can be synthesized in one step from Togni reagent. It has been used to achieve the first asymmetric trifluoromethoxylation reaction, realizing internal trifluoromethoxylation selectivity of alkenes (Scheme 1a).[6] Allenes exist in many natural products and pharmaceuticals.[7] The high activity of allenes makes them widely used as synthons for cyclization,[8] hydrogenheteroatom reactions[9] and multifunctionalization.[10] With our continuous efforts on the trifluoromethoxylation reactions,[11] we speculated that if allenes were subjected to our reaction system, OCF_3 would simultaneously be easily introduced into the products. However, the reaction of allenes presented more complex reactivity than that of alkenes. The versatile reactivities of allenes allow for the selective modification of different positions, enabling the synthesis of different regioselectivity functionalized compounds,[12] which could deliver more possible trifluoromethoxylation products. This interesting but questionable selectivity motivated us to investigate this chemistry. Conversely, due to the toxic reagents and the cost of transition metal catalysts or metallic reagents, transition metal-free processes with readily available OCF_3 sources have always attracted the attention of chemists. Herein, we report our efforts to develop a transition metal-free process for bromotrifluoromethoxylation of allenes with our trifluoromethoxy reagent (Scheme 1b).

Results and discussion

To validate our hypothesis, the initial study focused on the reaction of substrate **1a**, which was treated with the previous



Scheme 1 Intermolecular bromotrifluoromethoxylations.

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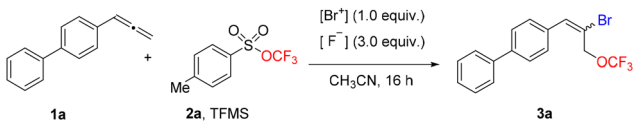
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olefin reaction conditions. We were delighted to find the bromosubstituted allyl trifluoromethoxy product **3a** in 24% yield with excellent regioselectivity but poor stereoselectivity (see Table S1 in the ESI†). We did not detect the benzyl trifluoromethoxylation product, similar to the olefin reactions in our previous work. Meanwhile, **3a** was obtained in 11% yield without the addition of silver fluoride and the ligand (Table 1, entry 1). NBS delivered the product in 16% yield and resulted in 56 : 44 *E*/*Z* selectivity. Other bromination reagents, such as TBCA, PHBP, TBCO and *N*-bromosaccharin afforded the desired product in less than 5% yield (Table 1, entries 3–6). Further optimization of the bromination reagents revealed that TBATB provided the best results, giving the corresponding product **3a** in 86% yield with 84 : 16 *E*/*Z* selectivity (Table 1, entry 7). The product **3a** was obtained with 76/24 *E*/*Z* selectivity when using 2 equivalents of TBATB (Table 1, entry 8). **3a** cannot be obtained when using NaF or CaF_2, perhaps due to the poor solubility of these two fluoride salts (Table 1, entries 9 and 11). Otherwise, KF can afford **3a** in 63% yield (Table 1, entry 10). However, the yield significantly decreased to 27% using silver fluoride (Table 1, entry 12). When the reaction was performed at 40 °C, the conversion after 16 h was slightly reduced (Table 1, entry 13). Meanwhile, upon heating the reaction mixture to 60 °C, compound **3a** was isolated in 78% yield (Table 1, entry 14).

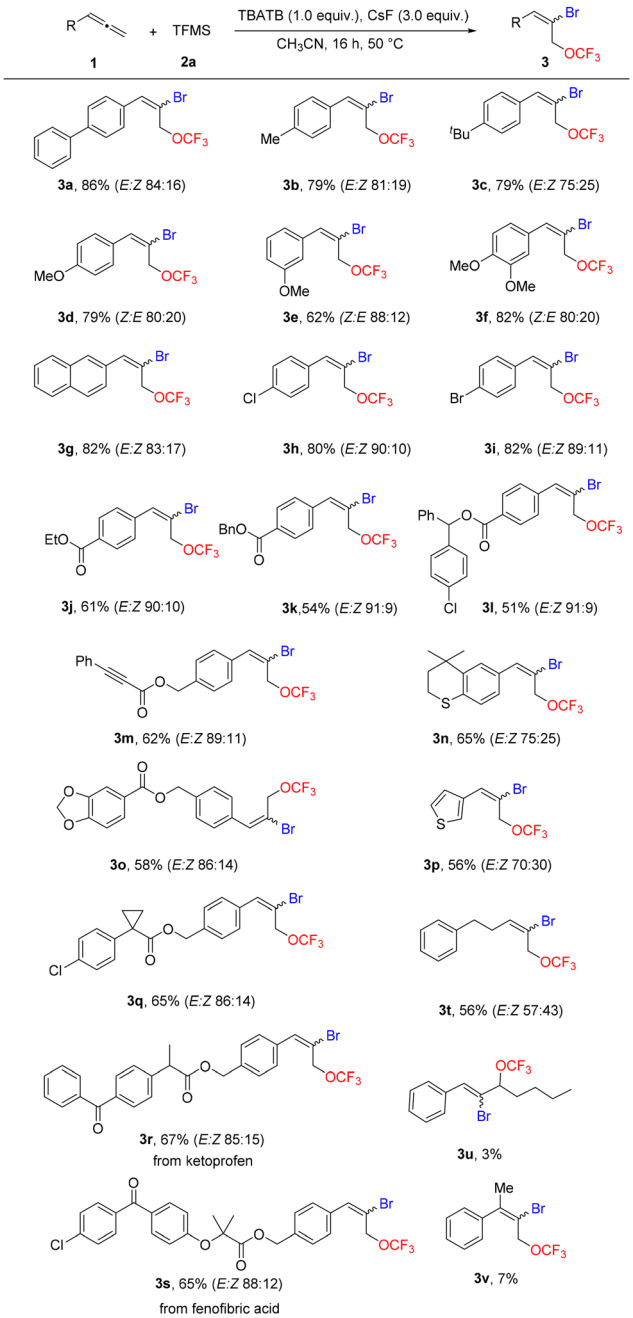
With the optimized conditions in hand (entry 7 in Table 1), we investigated the scope of the present transformation of allenes (Scheme 2). Generally, electron-donating groups

Table 1 Optimization of the reaction conditions[a]



Entry	[Br$^+$]	[F$^-$]	*T*/°C	Yield/%	*E*/*Z*
1	DBDMH	CsF	50	11	63/37
2	NBS	CsF	50	16	56/44
3	TBCA	CsF	50	1	ND
4	PHBP	CsF	50	0	—
5	TBCO	CsF	50	0	—
6	*N*-Bromosaccharin	CsF	50	2	50/50
7	TBATB	CsF	50	86	84/16
8[b]	TBATB	CsF	50	80	76/24
9	TBATB	NaF	50	0	—
10	TBATB	KF	50	63	78/22
11	TBATB	CaF_2	50	0	—
12	TBATB	AgF	50	27	52/48
13	TBATB	CsF	40	74	80/20
14	TBATB	CsF	60	78	77/23

[a] Reaction conditions: **1a** (0.1 mmol), TFMS (3 equiv.), [Br$^+$] (1 equiv.), [F$^-$] (3.0 equiv.), CH_3CN (0.25 mL), 16 h. Yields were determined by ^{19}FNMR spectroscopy with benzotrifluoride as a standard. [b] TBATB (2 equiv.). TBATB = tetrabutylammonium tribromide. DBDMH = 1,3-dibromo-5,5-dimethylhydantoin. NBS = *N*-bromosuccinimide. TBCA = 1,3,5-tribroMo-1,3,5-triazinane-2,4,6-trione. PHBP = pyridinium bromide perbromide. TBCO = 2,4,4,6-tetrabromo-2,5-cyclohexadien-1-one.



Scheme 2 Substrate scope of trifluoromethoxylation. Reaction conditions: **1** (0.2 mmol), TFMS (3.0 equiv.), TBATB (1.0 equiv.), CsF (3.0 equiv.), CH_3CN (0.5 mL), 50 °C, 16 h. Yields of isolated products are given, unless otherwise noted.

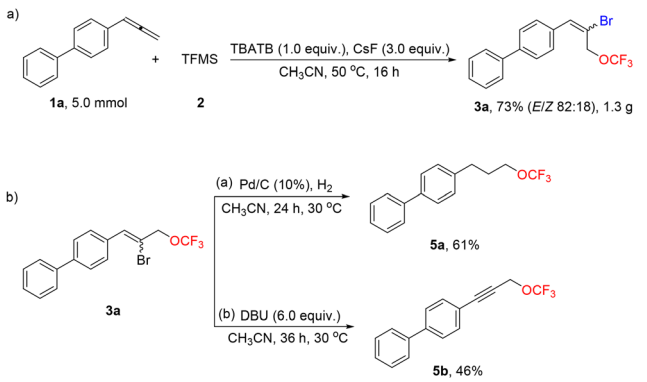
including phenyl, Me, ^{t}Bu, and OMe were well tolerated, providing the corresponding products in 62–86% yields (**3a–3f**). Naphthylallene was also reactive, giving a higher yield (**3g**, 82%, *E*/*Z* = 83 : 17). Substrates that contain a halogen can be applied to this system, which allowed further decoration (**3h** and **3i**, 80–82%, *E*/*Z* = 90 : 10 and 89 : 11). Allene substrates with an ester group were transformed into products **3j–3l** with good *E*-isomer selectivity, probably due to the electron-with-

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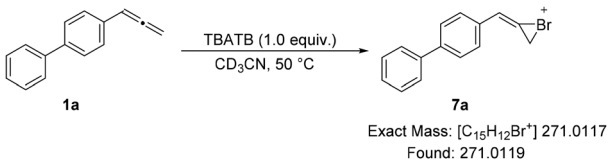
drawing induction effect. Moreover, alkyne can survive in these conditions (**3m**, 62%, *E*/*Z* = 89 : 11). The reaction of sulfur-containing heterocyclic allene **1n** delivered the product with a lower stereoselectivity of the C=C configuration (**3n**, 65%, *E*/*Z* = 75 : 25). In contrast, substrate **1o** with an oxygen-containing heterocyclic structure afforded the product in a better stereoselectivity (**3o**, 58%, *E*/*Z* = 86 : 14). Heteroaromatic allenes, such as thiopheneallene, were also suitable for this reaction (**3p**, 56%, *E*/*Z* = 70 : 30). Notably, a substrate containing a ternary ring was also compatible with this catalytic system (**3q**, 65%, *E*/*Z* = 86 : 14). The desired products **3r** and **3s** were obtained in moderate yields, which highlighted the good potential post-modification of this methodology for natural complex products. The product **3t** was observed using alkylallene as a substrate; moreover, the small steric hindrance led to a decrease in stereoselectivity. Internal allene **1u** and disubstituted allene **1v** afforded a small amount of the products, possibly due to the low activity of the substrates. Similarly, no product was obtained for the diarylallene, and the alkylallene gave the product in 20% yield (see Scheme S1 in the ESI†).

To evaluate the practicability and efficiency of this reaction, a gram-scale synthesis of **3a** was conducted under the standard reaction conditions in 73% yield (Scheme 3a). Further transformation of the bromoallylic trifluoromethoxy products for the construction of a series of OCF_3-containing compounds was investigated (Scheme 3b). Under hydrogenation conditions, alkyl trifluoromethoxy **5a** can be obtained with a yield of 61%. Propargyl trifluoromethoxy compound **5b** was produced with a yield of 46% by eliminating hydrogen bromide.

To gain more insight into the reaction mechanism, we carried out some control experiments (see Schemes S2 and S3 in the ESI†). Monitoring of the reaction between **1a** and TBATB by ^{1}H NMR spectrum showed that the characteristic hydrogen of the allene has undergone a shift. The HRMS-ESI (*m*/*z*) results indicated that alkene bromide ions were generated in the reaction (Scheme 4). This indicates the appearance of bromonium active intermediates. Meanwhile, we also conducted D-NOESY NMR spectra of **3a** and **3d** to confirm the configuration (see Schemes S4–S6 in the ESI†).[13]



Scheme 3 Gram-Scale synthesis and synthetic transformations.



Scheme 4 Mechanistic study.

Conclusions

We have presented the first example of bromotrifluoromethoxylation of allenes using TFMS as an OCF_3 source. By the combination of TBATB with CsF, a transition metal-free process with a readily available OCF_3 source provided a range of allylic trifluoromethoxy derivatives in up to 86% yield and with complete regiocontrol, which are difficult to access by other methods. Due to the easy decomposition of the OCF_3 group, carefully selecting the appropriate bromine source, reaction temperature, solvent, and base were very critical, as all the factors jointly guaranteed the smooth transformation. This discovery also expanded the scope of the application of the trifluoromethoxy reagent.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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